Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015119**	**Course Name:**	**Agile SoftwareDevelopment& DevOps Lab**

Name of Student:	**Chirag Raut**
Roll No. :	**85**
Experiment No.:	**07**
Title of the Experiment:	**To deploy a serverless function using AWS Lambda.**
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : **Ms. Kshitija Gharat**

Signature :

Date :

Aim : To deploy a serverless function using AWS Lambda by creating, configuring, testing, and executing a Lambda function in the AWS Cloud environment.

Objectives:

1. To understand the concept of serverless computing.
2. To understand the working of AWS Lambda.
3. To create and configure a Lambda function.
4. To write and deploy code using AWS Lambda.
5. To test the Lambda function using sample event data.
6. To understand event-driven execution.
7. To monitor Lambda function execution using AWS CloudWatch.
8. To understand the complete workflow of creating, deploying, and executing a serverless function.

Theory :

1. **Serverless Computing :** Serverless computing is a cloud computing model in which developers can run application code without managing servers or other underlying infrastructure. The cloud provider automatically manages resource allocation, scaling, availability, and maintenance. Developers can therefore focus on writing and deploying application code.

2. **AWS Lambda :** AWS Lambda is a serverless computing service provided by Amazon Web Services (AWS). It allows developers to execute code without provisioning or managing servers. A Lambda function runs whenever it is triggered by an event and automatically uses the required computing resources.

   The basic Lambda workflow is:

Event → Lambda Function → Code Execution → Response

Lambda supports programming languages such as Python, Java, JavaScript/Node.js, C#, Go, and Ruby.

3. **Lambda Function :** A Lambda function is a small piece of code designed to perform a specific task. It contains the application logic that is executed when the function is invoked.

For example, a simple Python Lambda function is:

```
1.  def lambda_handler(event, context):
2.  return {
3.  "statusCode": 200,
4.  "body": "Hello from AWS Lambda!"
5.  }
```

Here, event contains information about the event that triggered the function, while context provides information about the execution environment.

4. **Event-Driven Execution :** AWS Lambda follows an event-driven architecture. A function can be automatically invoked when an event occurs. Events may come from services such as Amazon S3, API Gateway, Amazon SQS, DynamoDB, or Amazon EventBridge.

For example, a Lambda function can be triggered when a file is uploaded to an S3 bucket or when an API request is received.

5. **Deployment and Execution :** To deploy a Lambda function, the developer first creates a function in the AWS Lambda console, selects the required runtime, writes or uploads the code, and configures the required permissions. The function can then be deployed and tested using a sample event.

The general process is:

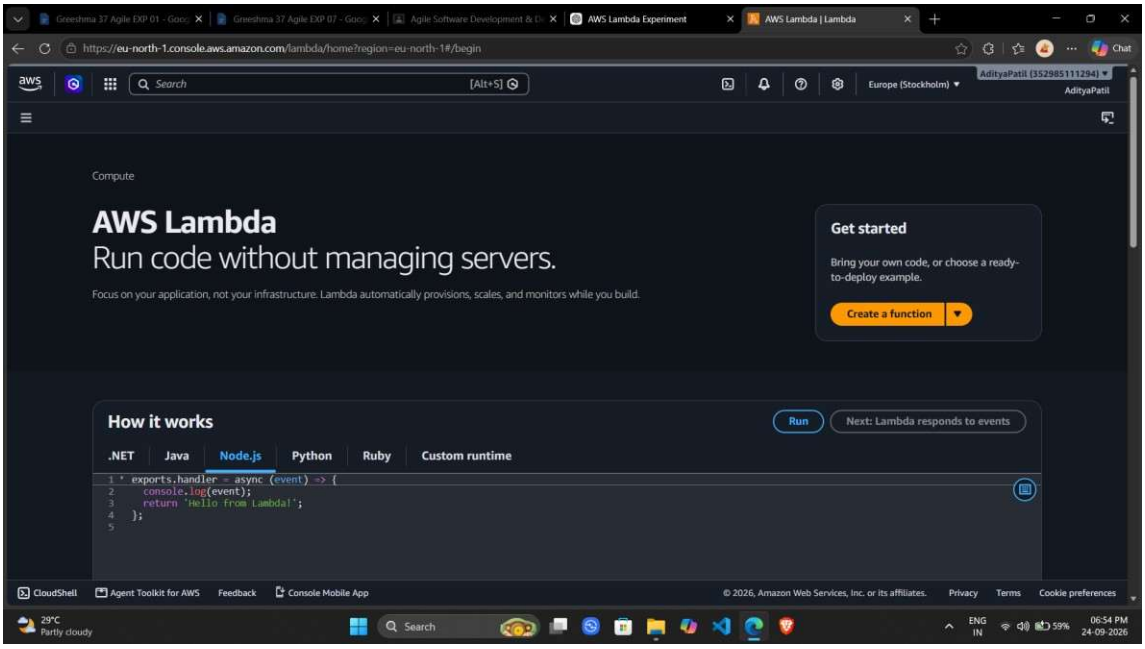
Create Function → Select Runtime → Write Code → Configure → Deploy → Test → Monitor

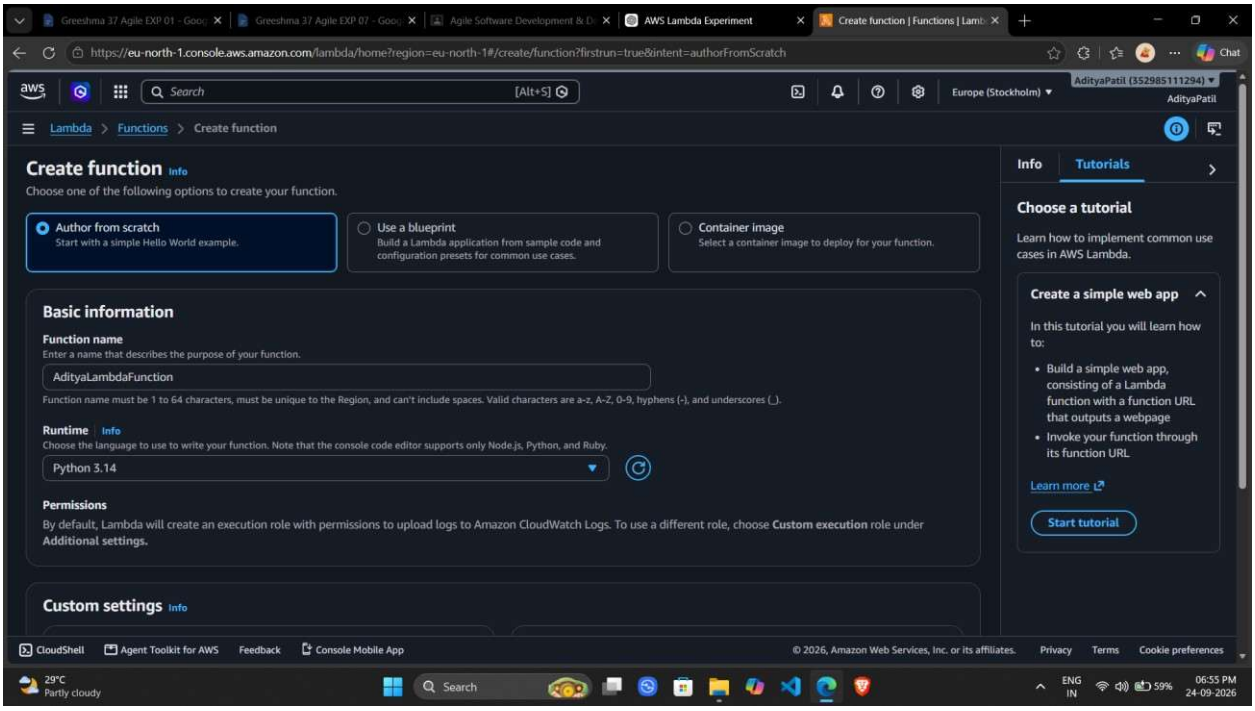
6. **Monitoring :** AWS Lambda integrates with Amazon CloudWatch for monitoring and logging. CloudWatch records execution logs and provides information such as execution duration, errors, and other performance metrics. These logs help developers identify problems and monitor the function.

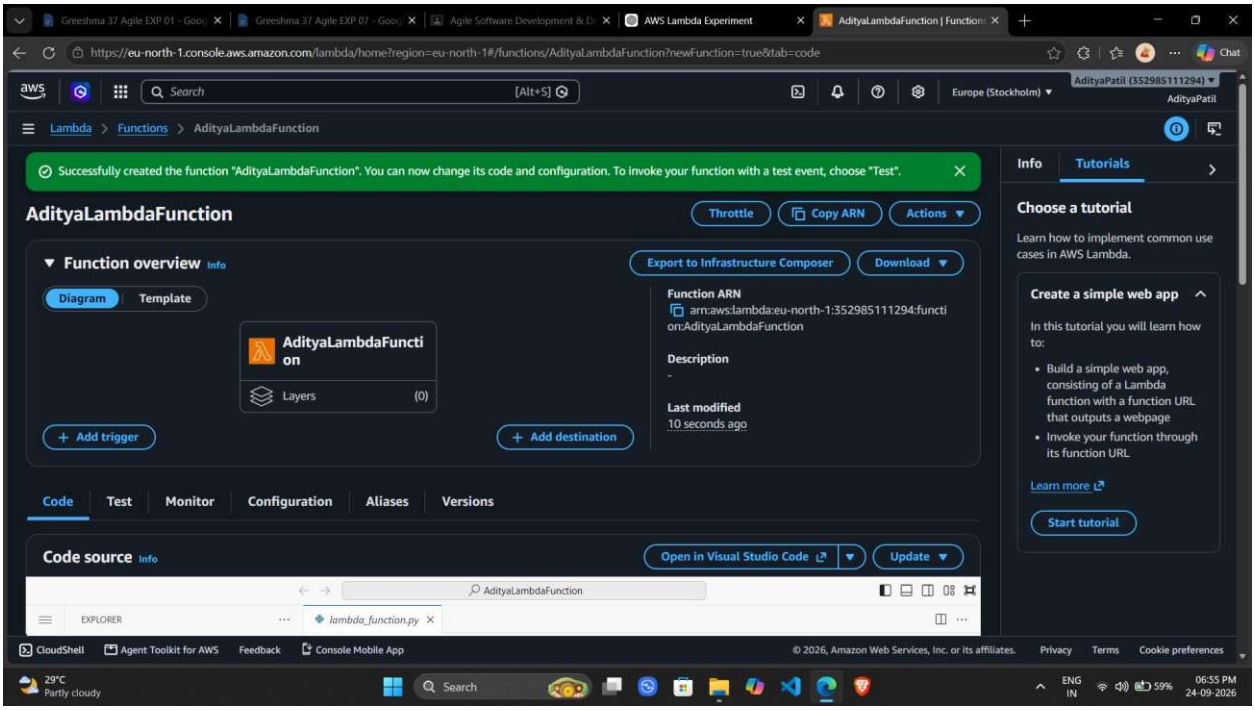
7. **Advantages of AWS Lambda :** The major advantages of AWS Lambda include:

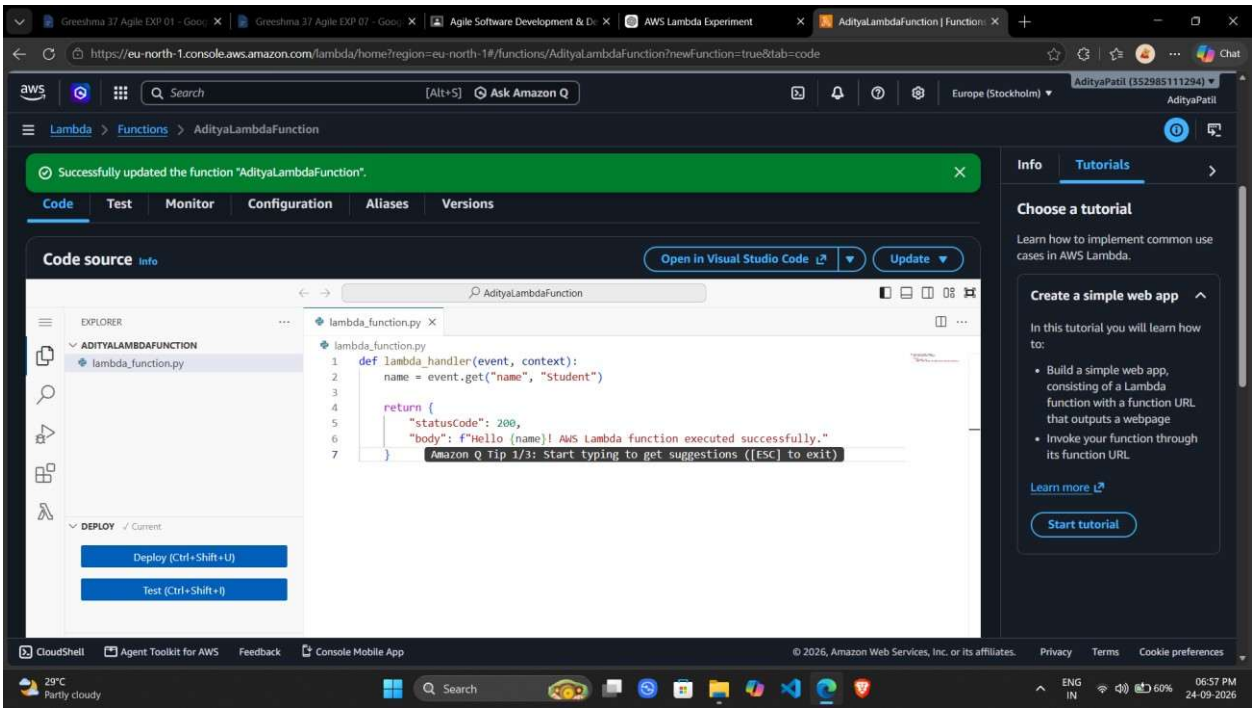
   - No server management is required.
   - Automatic scaling according to workload.
   - Pay-per-use resource model.
   - Easy deployment and testing.
   - Event-driven execution.
   - Integration with various AWS services.
   - High availability and reduced infrastructure management.

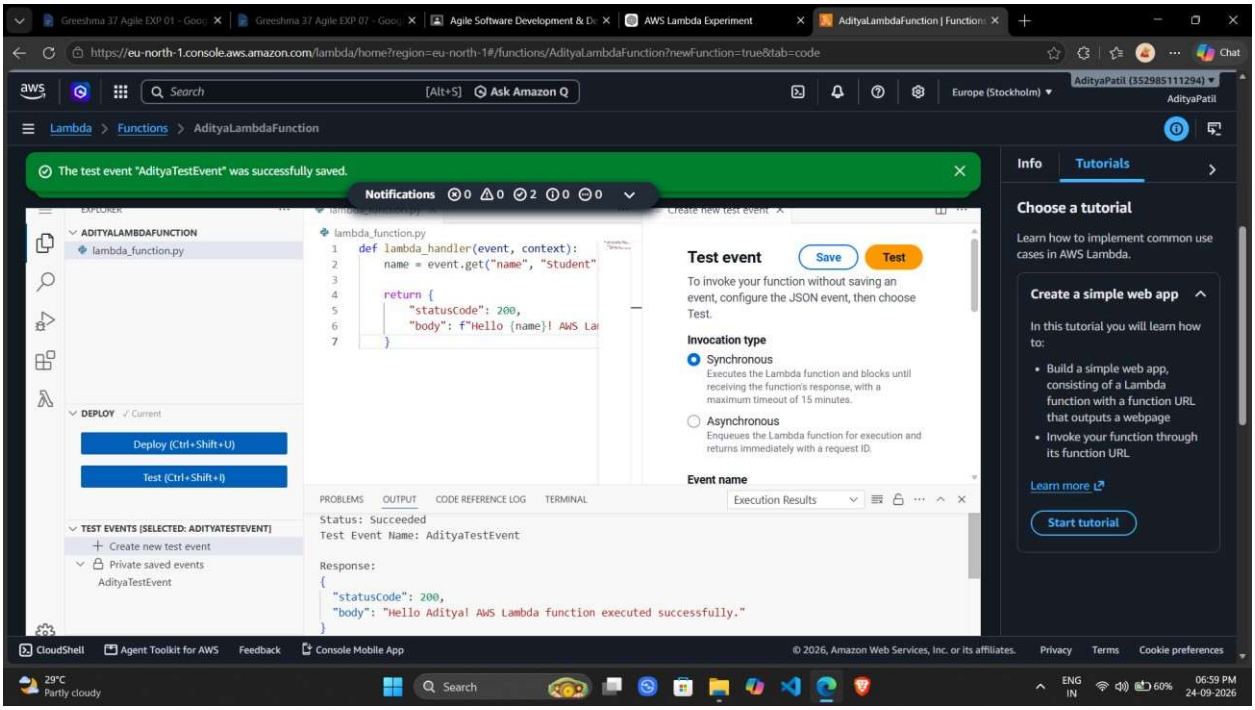
   - Thus, AWS Lambda provides a simple and scalable platform for developing and deploying serverless applications without directly managing servers. **Output:**

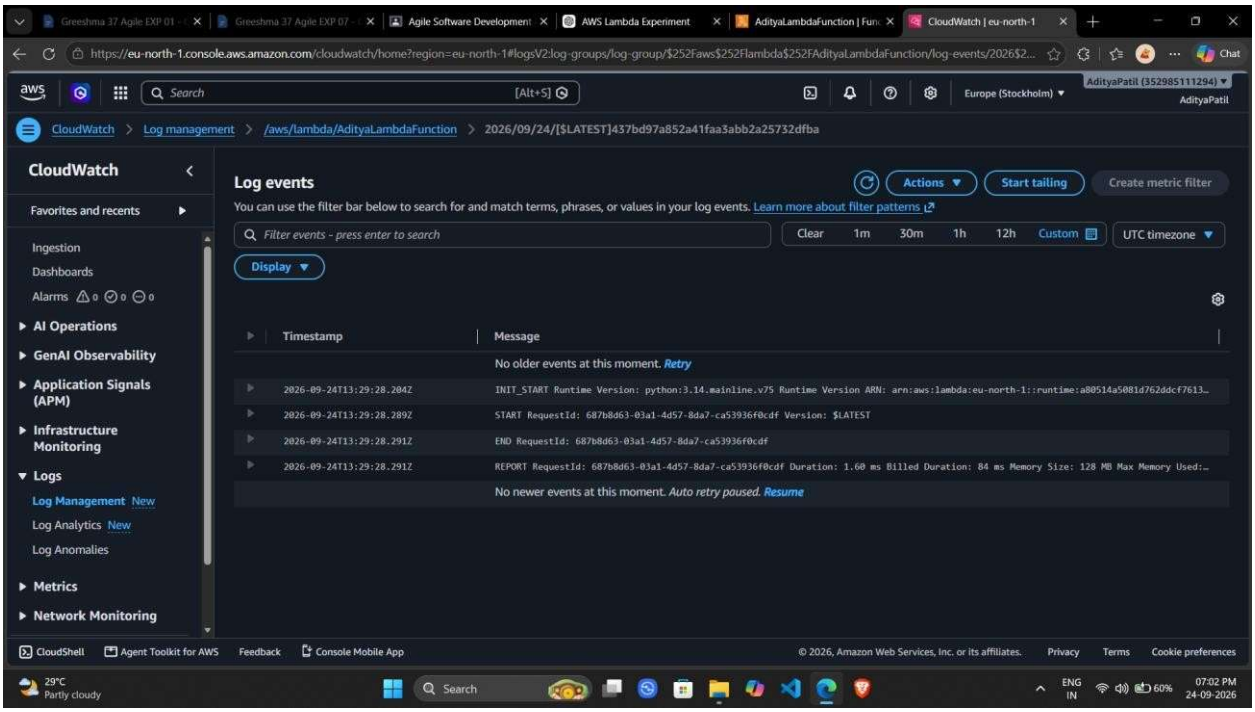

AWS Lambda Experiment
AWS Lambda | Lambda
https://eu-north-1.console.aws.amazon.com/lambda/home?region=eu-north-1#/begin
Search
[Alt+S]
Europe (Stockholm)
AdityaPatil
Compute
AWS Lambda
Run code without managing servers.
Focus on your application, not your infrastructure. Lambda automatically provisions, scales, and monitors while you build.
Get started
Bring your own code, or choose a ready-to-deploy example.
Create a function
How it works
Run
Next: Lambda responds to events
.NET
Java
Node.js
Python
Ruby
Custom runtime
exports.handler = async (event) => {
console.log(event);
return 'Hello from Lambda!';
};
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
29°C
Partly cloudy
Search
ENG
IN
06:54 PM
24-09-2026


AWS Lambda Experiment
Create function | Functions | Lamb
https://eu-north-1.console.aws.amazon.com/lambda/home?region=eu-north-1#/create/function?firstrun=true&intent=authorFromScratch
Search
[Alt+S]
Europe (Stockholm)
AdityaPatil
Lambda > Functions > Create function
Create function Info
Choose one of the following options to create your function.
Author from scratch
Start with a simple Hello World example.
Use a blueprint
Build a Lambda application from sample code and configuration presets for common use cases.
Container image
Select a container image to deploy for your function.
Basic information
Function name
Enter a name that describes the purpose of your function.
AdityaLambdaFunction
Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).
Runtime Info
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.
Python 3.14
Permissions
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose Custom execution role under Additional settings.
Custom settings Info
Info
Tutorials
Choose a tutorial
Learn how to implement common use cases in AWS Lambda.
Create a simple web app
In this tutorial you will learn how to:
Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
Invoke your function through its function URL
Learn more
Start tutorial
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
29°C
Partly cloudy
Search
ENG
IN
06:55 PM
24-09-2026


Successfully created the function "AdityaLambdaFunction". You can now change its code and configuration. To invoke your function with a test event, choose "Test".
AdityaLambdaFunction
Throttle
Copy ARN
Actions
Function overview
Export to Infrastructure Composer
Download
Diagram
Template
AdityaLambdaFunction
Layers (0)
+ Add trigger
+ Add destination
Function ARN
arn:aws:lambda:eu-north-1:352985111294:function:AdityaLambdaFunction
Description
Last modified
10 seconds ago
Code
Test
Monitor
Configuration
Aliases
Versions
Code source
Open in Visual Studio Code
Update
Info
Tutorials
Choose a tutorial
Learn how to implement common use cases in AWS Lambda.
Create a simple web app
In this tutorial you will learn how to:
Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
Invoke your function through its function URL
Learn more
Start tutorial


Successfully updated the function "AdityaLambdaFunction".
Code
Test
Monitor
Configuration
Aliases
Versions
Code source
Open in Visual Studio Code
Update
EXPLORER
ADITYALAMBDAFUNCTION
lambda_function.py
def lambda_handler(event, context):
name = event.get("name", "Student")
return {
"statusCode": 200,
"body": f"Hello {name}! AWS Lambda function executed successfully."
}
Amazon Q Tip 1/3: Start typing to get suggestions ([ESC] to exit)
DEPLOY
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
Info
Tutorials
Choose a tutorial
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Learn more
Start tutorial


The test event "AdityaTestEvent" was successfully saved.
Notifications
lambda_function.py
def lambda_handler(event, context):
name = event.get("name", "Student")
return {
"statusCode": 200,
"body": f"Hello {name}! AWS La
}
Test event
Save
Test
To invoke your function without saving an event, configure the JSON event, then choose Test.
Invocation type
Synchronous
Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes.
Asynchronous
Enqueues the Lambda function for execution and returns immediately with a request ID.
Event name
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
Create new test event
Private saved events
AdityaTestEvent
Execution Results
Status: Succeeded
Test Event Name: AdityaTestEvent
Response:
{
"statusCode": 200,
"body": "Hello Aditya! AWS Lambda function executed successfully."
}
Info
Tutorials
Choose a tutorial
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Start tutorial


CloudWatch
Log events
Actions
Start tailing
Create metric filter
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Filter events - press enter to search
Clear
1m
30m
1h
12h
Custom
UTC timezone
Display
Timestamp
Message
No older events at this moment. Retry
2026-09-24T13:29:28.204Z INIT_START Runtime Version: python:3.14.mainline.v75 Runtime Version ARN: arn:aws:lambda:eu-north-1::runtime:a80514a5081d762ddcf7613…
2026-09-24T13:29:28.289Z START RequestId: 687b8d63-03a1-4d57-8da7-ca53936f0cdf Version: $LATEST
2026-09-24T13:29:28.291Z END RequestId: 687b8d63-03a1-4d57-8da7-ca53936f0cdf
2026-09-24T13:29:28.291Z REPORT RequestId: 687b8d63-03a1-4d57-8da7-ca53936f0cdf Duration: 1.60 ms Billed Duration: 84 ms Memory Size: 128 MB Max Memory Used:…
No newer events at this moment. Auto retry paused. Resume
Favorites and recents
Ingestion
Dashboards
Alarms
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management New
Log Analytics New
Log Anomalies
Metrics
Network Monitoring

Conclusion:

The serverless function was successfully created, configured, deployed, and tested using AWS Lambda. The function was executed successfully using a test event, and its execution was verified through Amazon CloudWatch logs. This experiment demonstrated how AWS Lambda enables serverless and event-driven application execution without managing servers.

Questions to be answered by students

1. What is AWS Lambda?

Answer: AWS Lambda is a serverless computing service that allows users to run code in response to events without managing servers or infrastructure.

2. What is the purpose of Amazon CloudWatch in AWS Lambda?

Answer: Amazon CloudWatch is used to monitor Lambda functions and view execution logs, duration, memory usage, and errors.